Uniwersytet SWPS

Task list no. 6 – In this lab, we will work with a Max-Priority queue and implement the non-comparison sorting algorithm, counting sort.

Task 1. Use the MAX-HEAPIFY procedure to build a Max-Priority Queue that supports the following operations:

- HEAP-MAXIMUM: returns the maximum element in the priority queue.
- HEAP-EXTRACT-MAX: removes and returns the maximum element from the priority queue.
- HEAP-INCREASE-KEY: increases the value of a given key in the priority queue.
- MAX-HEAP-INSERT: inserts a new key into the priority queue.

1-1: Insert the elements (15, 10, 30, 40, 5) into a priority queue using the MAX-HEAP-INSERT operation. After each insertion, print the resulting heap.
1-2: Find the maximum element of the heap using HEAP-MAXIMUM, and print the returned value.
1-3: Extract the maximum by calling HEAP-EXTRACT-MAX. Print the current state of the heap.
1-4: Increase the key of the third element in the heap to 35 using HEAP-INCREASE-KEY, print the current state of the heap.
1-5 (homework) **:** Implement HEAP-DECREASE-KEY(A, i, new_key) that decreases the key of the element at index i. You must use MAX-HEAPIFY in this function. Decrease the fourth element of your heap, and print the current state of the heap.

Task 2. Implement counting sort.

2-1: Run the counting sort on the array A = {2,5,3,0,2,3,0,3} and print your array after sorting.

Note. The lab report should contain your code along with the results.

Note. ALL PROGRAMS SHOULD COMPILE WITHOUT ERROR AND WORK (i.e. DO EXACTLY WHAT IS DESCRIBED IN THE TASK)!!!

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